1 Basic concepts

1.1 What is a quantum phase transition?

Consider a Hamiltonian, $H(g)$, whose degrees of freedom reside on the sites of a lattice, and which varies as a function of a dimensionless coupling, g. Let us follow the evolution of the ground state energy of $H(g)$ as a function of g. For the case of a finite lattice, this ground state energy will generically be a smooth, analytic function of g. The main possibility of an exception comes from the case when g couples only to a conserved quantity (i.e. $H(g) = H_0 + gH_1$, where H_0 and H_1 commute). This means that H_0 and H_1 can be simultaneously diagonalized and so the eigenfunctions are independent of g even though the eigenvalues vary with g; then there can be a level-crossing where an excited level becomes the ground state at $g = g_c$ (say), creating a point of nonanalyticity of the ground state energy as a function of g (see Fig. 1.1). The possibilities for an *infinite* lattice are richer. An avoided level-crossing between the ground and an excited state in a finite lattice could become progressively sharper as the lattice size increases, leading to a nonanalyticity at $g = g_c$ in the infinite lattice limit. We shall identify any point of nonanalyticity in the ground state energy of the infinite lattice system as a quantum phase transition: The nonanalyticity could be either the limiting case of an avoided level-crossing or an actual level-crossing. The first kind is more common, but we shall also discuss transitions of the second kind in Chapters 16 and 19. The phase transition is usually accompanied by a qualitative change in the nature of the correlations in the ground state, and describing this change will clearly be one of our major interests.

Actually our focus will be on a limited class of quantum phase transitions – those that are *second order*. Loosely speaking, these are transitions at which the characteristic energy scale of fluctuations above the ground state vanishes as g approaches g_c. Let the energy Δ represent a scale characterizing some significant spectral density of fluctuations at zero temperature (T) for $g \neq g_c$. Thus Δ could be the energy of the lowest excitation above the ground state, if this is nonzero (i.e. there is an energy gap Δ), or if there are excitations at arbitrarily low energies in the infinite lattice limit (i.e. the energy spectrum is *gapless*), Δ is the scale at which there is a qualitative change in the nature of the frequency spectrum from its lowest frequency to its higher frequency behavior. In most cases, we will find that as g approaches g_c, Δ vanishes as

$$\Delta \sim J|g - g_c|^{z\nu}, \tag{1.1}$$

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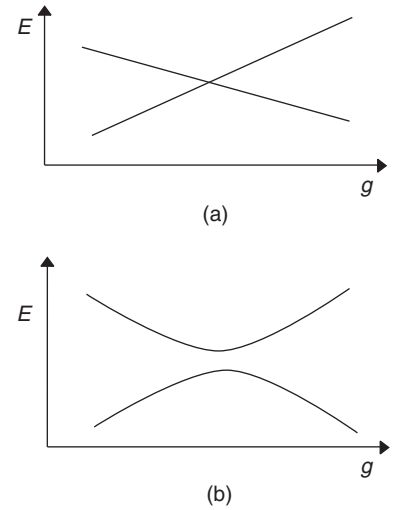

Fig. 1.1 Low eigenvalues, E, of a Hamiltonian $H(g)$ on a finite lattice, as a function of some dimensionless coupling, g. For the case where $H(g) = H_0 + gH_1$, where H_0 and H_1 commute and are independent of g, there can be an actual level-crossing, as in (a). More generally, however, there is an "avoided level-crossing," as in (b).

(exceptions to this behavior appear in Section 20.2.6). Here J is the energy scale of a characteristic microscopic coupling, and $z\nu$ is a *critical exponent*. The value of $z\nu$ is usually *universal*, that is, it is independent of most of the microscopic details of the Hamiltonian $H(g)$ (we shall have much more to say about the concept of universality below, and in the following chapters). The behavior (1.1) holds both for $g > g_c$ and for $g < g_c$ with the same value of the exponent $z\nu$, but with different nonuniversal constants of proportionality. We shall sometimes use the symbol Δ_+ (Δ_-) to represent the characteristic energy scale for $g > g_c$ ($g < g_c$).

In addition to a vanishing energy scale, second-order quantum phase transitions invariably have a diverging characteristic length scale ξ. This could be the length scale determining the exponential decay of equal-time correlations in the ground state or the length scale at which some characteristic crossover occurs to the correlations at the longest distances. This length diverges as

$$\xi^{-1} \sim \Lambda |g - g_c|^{\nu}, \tag{1.2}$$

where ν is a critical exponent, and Λ is an inverse length scale (a "momentum cutoff") of order the inverse lattice spacing. The ratio of the exponents in (1.1) and (1.2) is z, the dynamic critical exponent. The characteristic energy scale vanishes as the zth power of the characteristic inverse length scale

$$\Delta \sim \xi^{-z}. \tag{1.3}$$

It is important to note that the discussion above refers to singularities in the *ground state* of the system. So strictly speaking, quantum phase transitions occur only at zero temperature, $T = 0$. Because all experiments are necessarily at some nonzero, though

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possibly very small, temperature, a central task of the theory of quantum phase transitions is to describe the consequences of this $T = 0$ singularity on physical properties at $T > 0$. It turns out that working outward from the quantum critical point at $g = g_c$ and $T = 0$ is a powerful way of understanding and describing the thermodynamic and dynamic properties of numerous systems over a broad range of values of $|g - g_c|$ and T. Indeed, it is not even necessary that the system of interest ever have its microscopic couplings reach a value such that $g = g_c$: it can still be very useful to argue that there is a quantum critical point at a physically inaccessible coupling $g = g_c$ and to develop a description in the deviation $|g - g_c|$. It is one of the purposes of this book to describe the physical perspective that such an approach offers, and to contrast it with more conventional expansions about very weak (say $g \to 0$) or very strong couplings (say $g \to \infty$).

1.2 Nonzero temperature transitions and crossovers

Let us now discuss some basic aspects of the $T > 0$ phase diagram. First, let us ask only about the presence of phase transitions at nonzero T. With this limited criterion, there are two important possibilities for the $T > 0$ phase diagram of a system near a quantum critical point. These are shown in Fig. 1.2, and we will meet examples of both kinds in this book. In the first, shown in Fig. 1.2a, the thermodynamic singularity is present only at $T = 0$, and all $T > 0$ properties are analytic as a function of g near $g = g_c$. In the second, shown in

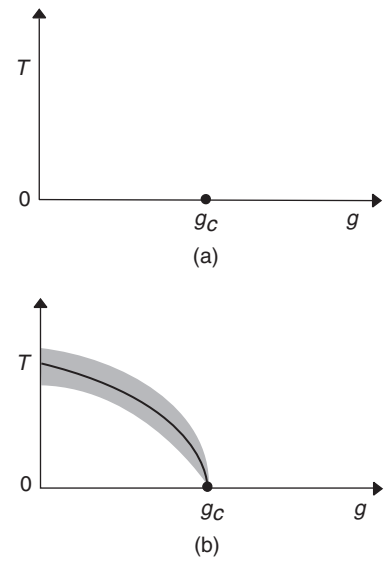

Fig. 1.2 Two possible phase diagrams of a system near a quantum phase transition. In both cases there is a quantum critical point at $g = g_c$ and $T = 0$. In (b), there is a line of $T > 0$ second-order phase transitions terminating at the quantum critical point. The theory of phase transitions in classical systems driven by thermal fluctuations can be applied within the shaded region of (b).

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Fig. 1.2b, there is a line of $T > 0$ second-order phase transitions (this is a line at which the thermodynamic free energy is not analytic) that terminates at the $T = 0$ quantum critical point at $g = g_c$.

Moving beyond phase transitions, let us ask some basic questions about the dynamics of the system. A very general way to characterize the dynamics at $T > 0$ is in terms of the thermal equilibration time τ_{eq}. This is the characteristic time in which *local* thermal equilibrium is established after imposition of a weak external perturbation (say, a heat pulse). Here we are excluding equilibration with respect to globally conserved quantities (such as energy or charge) which will take a long time to equilibrate, dependent upon the length scale of the perturbation: hence the emphasis on local equilibration. Global equilibration is described by the equations of hydrodynamics, and we expect such equations to apply in all cases at times much larger than τ_{eq}. We focus here on the value of τ_{eq} as a function of $g - g_c$ and T. From the energy scales discussed in Section 1.1, we can immediately draw an important distinction between two regimes of the phase diagram. We characterized the ground state by the energy Δ in (1.1). At nonzero temperature, we have a second energy scale, $k_B T$. Comparing the values of Δ and $k_B T$, we are immediately led to the important phase diagram in Fig. 1.3. We will see that the two regimes, $\Delta > k_B T$ and $\Delta < k_B T$, are distinguished by different theories of thermal equilibration and of the values of τ_{eq}. In the regime where $\Delta > k_B T$, we will always find long equilibration times which satisfy

$$\tau_{\text{eq}} \gg \frac{\hbar}{k_B T}, \quad \Delta > k_B T. \tag{1.4}$$

One of the important consequences of this large value of τ_{eq} is that the dynamics of the system becomes effectively classical. Thus we can use classical equations of motion to describe the re-equilibration dynamics at the time scale τ_{eq}.

Let us now turn our attention to the important "Quantum Critical" region in Fig. 1.3, where $k_B T > \Delta$. We shall mainly be interested in quantum critical points which are strongly interacting, and not amenable to a nearly-free particle description. In such cases we find a short equilibration time given by

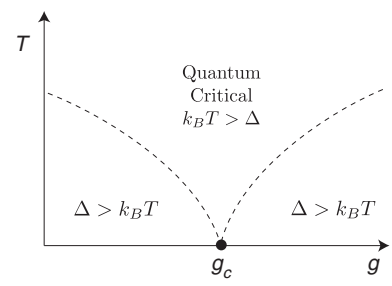

Fig. 1.3 Separation of the phase diagram into distinct regimes determined by the energy scale Δ, which characterizes the ground state, and $k_B T$. The dashed lines are not phase transitions, but smooth crossovers at $T \sim |g - g_c|^{z\nu}$. The phase transition in Fig. 1.2b lies within the $\Delta > k_B T$ region, and is not shown above.

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$$\tau_{\text{eq}} \sim \frac{\hbar}{k_B T}, \quad k_B T > \Delta. \tag{1.5}$$

Now the equilibration occurs in a time which is actually independent of the microscopic energy scale J, and is determined by $k_B T$ alone. Moreover, and most interestingly, we cannot use an effectively classical description for the re-equilibration at times of order τ_{eq}. Quantum and thermal fluctuations are equally important in the dynamics in the quantum critical region, and developing a theory for this dynamics will be a central focus of Part III.

What about the $T > 0$ phase transition line in Fig. 1.2b? We have not shown this line in Fig. 1.3. Such a transition should be viewed as reflecting the physics of the $\Delta > k_B T$ region, and so the transition line lies in the corresponding region of Fig. 1.3. In other words, this transition is not really a property of the quantum critical point at $g = g_c$, but of the quantum phase at $g < g_c$. (There could also be a separate transition reflecting the physics of the $g > g_c$ phase, which we have not shown in our phase diagrams.) As we move closer to this phase transition line, we will show that not only does τ_{eq} become long, but so do all the time scales associated with long wavelength thermal fluctuations. Indeed we will find that the typical frequency at which the important long-distance degrees of freedom fluctuate, ω_{typ}, satisfies

$$\hbar\omega_{\text{typ}} \ll k_B T. \tag{1.6}$$

Under these conditions, it will be seen that a purely *classical* description can be applied to these important degrees of freedom – this classical description works in the shaded region of Fig. 1.2b. Consequently, the ultimate critical singularity along the line of $T > 0$ phase transitions in Fig. 1.2b is described by the theory of second-order phase transitions in classical systems. This theory was developed thoroughly in the past three decades and has been explained in many popular reviews and books [59, 172, 244, 312, 557]. We will discuss the needed basic features of this theory in Chapters 3 and 4. Note that the shaded region of classical behavior in Fig. 1.2b lies within the wider window of the phase diagram, with moderate values of $|g - g_c|$ and T, which we asserted above should be described as an expansion about the quantum critical point at $g = g_c$ and $T = 0$. So our study of quantum phase transitions will also apply to the shaded region of Fig. 1.2b, where it will yield information complementary to that available by directly thinking of the $T > 0$ phase transition in terms of purely classical models.

We note that phase transitions in classical models are driven only by thermal fluctuations, as classical systems usually freeze into a fluctuationless ground state at $T = 0$. In contrast, quantum systems have fluctuations driven by the Heisenberg uncertainty principle even in the ground state, and these can drive interesting phase transitions at $T = 0$. The $T > 0$ region in the vicinity of a quantum critical point therefore offers a fascinating interplay of effects driven by quantum and thermal fluctuations; sometimes, as in the shaded region of Fig. 1.2b, we can find some dominant, effective degrees of freedom whose fluctuations are purely classical and thermal, and then the classical theory will apply. However, as already noted, our attention will not be limited to such regions, and we shall be interested in a broader section of the phase diagram.

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1.3 Experimental examples

To make the concepts of the previous sections less abstract, let us mention some experimental studies of simple second-order quantum phase transitions. We will meet numerous other examples in this book, but for now we focus on examples directly related to the canonical theoretical models of quantum phase transitions to be discussed in Section 1.4, and in Parts II and III.

- The low-lying magnetic excitations of the insulator $LiHoF_4$ consist of fluctuations of the Ho ions between two spin states that are aligned parallel and antiparallel to a particular crystalline axis. These states can be represented by a two-state "Ising" spin variable on each Ho ion. At $T = 0$, the magnetic dipolar interactions between the Ho ions cause all the Ising spins to align in the same orientation, and so the ground state is a ferromagnet. Bitko, Rosenbaum, and Aeppli [49] placed this material in a magnetic field transverse to the magnetic axis. Such a field induces quantum tunneling between the two states of each Ho ion, and a sufficiently strong tunneling rate can eventually destroy the long-range magnetic order. Such a quantum phase transition was indeed observed [49], with the ferromagnetic moment vanishing continuously at a quantum critical point. Note that such a transition can, in principle, occur precisely at $T = 0$, when it is driven entirely by quantum fluctuations. We shall call the $T = 0$ state without magnetic order a *quantum paramagnet*. However, we can also destroy the magnetic order at a fixed transverse magnetic field (possibly zero), simply by raising the temperature, enabling the material to undergo a conventional Curie transition to a high-temperature magnetically disordered state. Among the objectives of this book is to provide a description of the intricate crossover between the zero-temperature quantum transition and the finite-temperature transition driven partially by thermal fluctuations; we shall also delineate the important differences between the $T = 0$ quantum paramagnet and the high-temperature "thermal paramagnet;" see Chapters 11, 13, and 14.

  A more recent realization of an Ising model in a transverse field has appeared in experiments by Coldea and collaborators [90] on crystals of $CoNb_2O_6$, which belongs to the columbite family of minerals. In this case, the Ising spin resides on the Co^{++} ion, again aligned by the spin–orbit interaction to orient parallel or anti-parallel to a crystalline axis. An important difference from $LiHoF_4$ is that the interactions between the spins are essentially nearest-neighbor, and the long-range dipolar couplings are unimportant; the short-range interactions arise from the Heisenberg exchange process, and their energy scale is determined by the electrostatic Coulomb interactions. Thus $CoNb_2O_6$ provides a nearly ideal realization of the quantum Ising models which will be the focus of our study in Parts II and III. The dominant exchange couplings are along a particular crystalline axis, and so it is also a useful testing ground for exact results in one dimension.
- Experiments on ultracold atoms in optical lattices by Greiner, Bloch, and collaborators [175] have provided a celebrated example of the superfluid–insulator quantum phase transition. Atoms of ^{87}Rb are cooled to temperatures so low that their quantum statistics

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is important. These atoms are bosons and so they ultimately Bose condense into a superfluid state. Then, by applying a periodic potential on the atoms by an optical lattice, Greiner *et al.* localized the atoms in the minima of the periodic potential, leading to a quantum phase transition to an insulating state. At densities where the number of atoms is commensurate with the number of minima of the periodic potential, this transition is described by the O(2) quantum rotor model, which we introduce in Section 1.4 and discuss at length in Parts II and III.

- $TlCuCl_3$ is an insulator whose only low-lying electronic excitations are rotations of the $S = 1/2$ spins residing on the Cu^{++} ions. Unlike the case for the Co^{++} ions in $CoNb_2O_6$, the spin-orbit interactions are relatively weak on Cu^{++}, and a single spin can freely orient along any direction in spin space. A special feature of the crystal structure of $TlCuCl_3$ is that the Cu atoms are naturally dimerized, i.e. each Cu site has a single partner Cu site, and the exchange interactions are strongest between the partners in each pair. The exchange interaction has an antiferromagnetic sign, and consequently neighboring spins prefer to be oriented in anti-parallel directions. Under ambient pressure, each Cu spin forms a singlet valence bond with its partner, much like that between the two electrons in a hydrogen molecule. Thus although the neighboring spins within a dimer are always anti-parallel, they fluctuate along all directions in spin space in a rotationally invariant manner. We will refer to this state as a quantum paramagnet; it has an energy gap to all excitations above the ground state. Under applied pressure, $TlCuCl_3$ undergoes a quantum phase transition [414] to an ordered antiferromagnet: a Néel state. In this Néel state, the spins freeze into a definite orientation so that nearby spins are anti-parallel to each other. Such an arrangement is more nearly optimal when the exchange couplings between spins in different dimers are significant. As we discuss below in Section 1.4, this transition between the quantum paramagnet and the Néel state is described by the O(3) quantum rotor model, which will also be discussed in Parts II and III.

1.4 Theoretical models

Our strategy in this book will be to thoroughly analyze the physical properties of quantum phase transitions in two simple theoretical model systems in Parts II and III: the quantum Ising and rotor models. Fortunately, these simple models also have direct experimental realizations in the systems already surveyed in Section 1.3. Below, we introduce the quantum Ising and rotor models in turn, discussing the nature of the quantum phase transitions in them, and relating them to the experimental systems above. Other experimental connections will be discussed in subsequent chapters.

Part IV will survey some important quantum phase transitions in other models of physical interest. Our motivation in dividing the discussion in this manner is mainly pedagogical: the quantum transitions of the Ising/rotor models have an essential simplicity, but their behavior is rich enough to display most of the basic phenomena we wish to explore. It will therefore pay to meet the central physical ideas in this simple context first.

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1.4.1 Quantum Ising model

We begin by writing down the Hamiltonian of the quantum Ising model. It is

$$H_I = -Jg \sum_i \hat{\sigma}_i^x - J \sum_{\langle ij \rangle} \hat{\sigma}_i^z \hat{\sigma}_j^z. \tag{1.7}$$

As in the general notation introduced above, $J > 0$ is an exchange constant, which sets the microscopic energy scale, and $g > 0$ is a dimensionless coupling, which will be used to tune H_I across a quantum phase transition. The quantum degrees of freedom are represented by operators $\hat{\sigma}_i^{z,x}$, which reside on the sites, i, of a hypercubic lattice in d dimensions; the sum $\langle ij \rangle$ is over pairs of nearest-neighbor sites i, j. The $\hat{\sigma}_i^{x,z}$ are the familiar Pauli matrices; the matrices on different sites i act on different spin states, and so matrices with $i \neq j$ commute with each other. In the basis where the $\hat{\sigma}_i^z$ are diagonal, these matrices have the well-known form

$$\hat{\sigma}^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \hat{\sigma}^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{1.8}$$

on each site i. We will denote the eigenvalues of $\hat{\sigma}_i^z$ simply by σ_i^z, and so σ_i^z takes the values ± 1. We identify the two states with eigenvalues $\sigma_i^z = +1, -1$ as the two possible orientations of an "Ising spin," which can be oriented up or down in $|\uparrow\rangle_i, |\downarrow\rangle_i$. Consequently at $g = 0$, when H_I involves only the $\hat{\sigma}_i^z$, H_I will be diagonal in the basis of eigenvalues of $\hat{\sigma}_i^z$, and it reduces simply to the familiar classical Ising model. However, the $\hat{\sigma}_i^x$ are off-diagonal in the basis of these states, and therefore they induce quantum-mechanical tunneling events that flip the orientation of the Ising spin on a site. The physical significance of the two terms in H_I should be clear in the context of our earlier discussion in Section 1.3 for $LiHoF_4$ and $CoNb_2O_6$. The term proportional to J is the magnetic interaction between the spins, which prefers their global ferromagnetic alignment. While the interaction in $LiHoF_4$ has a long-range dipolar nature, that in $CoNb_2O_6$ has a nearest-neighbor form like that in (1.7). The term proportional to Jg is the applied external transverse magnetic field, which disrupts the magnetic order.

Let us make these qualitative considerations somewhat more precise. The ground state of H_I can depend only upon the value of the dimensionless coupling g, and so it pays to consider the two opposing limits $g \gg 1$ and $g \ll 1$.

First consider $g \gg 1$. In this case the first term in (1.7) dominates, and, to leading order in $1/g$, the ground state is simply

$$|0\rangle = \prod_i |\rightarrow\rangle_i, \tag{1.9}$$

where

$$\begin{aligned} |\rightarrow\rangle_i &= (|\uparrow\rangle_i + |\downarrow\rangle_i)/\sqrt{2}, \\ |\leftarrow\rangle_i &= (|\uparrow\rangle_i - |\downarrow\rangle_i)/\sqrt{2}, \end{aligned} \tag{1.10}$$

are the two eigenstates of $\hat{\sigma}_i^x$ with eigenvalues ± 1. The values of σ_i^z on different sites are totally uncorrelated in the state (1.9), and so $\langle 0|\hat{\sigma}_i^z \hat{\sigma}_j^z|0\rangle = \delta_{ij}$. Perturbative corrections in

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$1/g$ will build in correlations in σ^z that increase in range at each order in $1/g$; for g large enough these correlations are expected to remain short-ranged, and we expect in general that

$$\langle 0|\hat{\sigma}_i^z \hat{\sigma}_j^z|0\rangle \sim e^{-|x_i - x_j|/\xi} \tag{1.11}$$

for large $|x_i - x_j|$, where x_i is the spatial coordinate of site i, $|0\rangle$ is the exact ground state for large-g, and ξ is the "correlation length" introduced above (1.2).

Next we consider the opposing limit $g \ll 1$. We will find that the nature of the ground state is qualitatively different from the large-g limit above, and we shall use this to argue that there must be a quantum phase transition between the two limiting cases at a critical $g = g_c$ of order unity. For $g \ll 1$, the second term in (1.7) coupling neighboring sites dominates; at $g = 0$ the spins are either all up or all down (in eigenstates of σ^z):

$$|\uparrow\rangle = \prod_i |\uparrow\rangle_i \quad \text{or} \quad |\downarrow\rangle = \prod_i |\downarrow\rangle_i \,. \tag{1.12}$$

Turning on a small-g will mix in a small fraction of spins of the opposite orientation, but in an infinite system the degeneracy will survive at any finite order in a perturbation theory in g. This is because there is an exact global Z_2 symmetry transformation (generated by the unitary operator $\prod_i \sigma_i^x$), which maps the two ground states into each other, under which H_I remains invariant:

$$\hat{\sigma}_i^z \to -\hat{\sigma}_i^z, \quad \hat{\sigma}_i^x \to -\hat{\sigma}_i^x, \tag{1.13}$$

and there is no tunneling matrix element between the majority up and down spin sectors of the infinite system at any finite order in g. The mathematically alert reader will note that establishing the degeneracy to all orders in g, is not the same thing as establishing its existence for any small nonzero g, but more sophisticated considerations show that this is indeed the case. A thermodynamic system will always choose one or other of the states as its ground states (which may be preferred by some infinitesimal external perturbation), and this is commonly referred to as a "spontaneous breaking" of the Z_2 symmetry. As in the large-g limit, we can characterize the ground states by the behavior of correlations of $\hat{\sigma}_i^z$; the nature of the states (1.12) and small-g perturbation theory suggest that

$$\lim_{|x_i - x_j| \to \infty} \langle 0|\hat{\sigma}_i^z \hat{\sigma}_j^z|0\rangle = N_0^2, \tag{1.14}$$

where $|0\rangle$ is either of the ground states obtained from $|\uparrow\rangle$ or $|\downarrow\rangle$ by perturbation theory in g, and $N_0 \neq 0$ is the "spontaneous magnetization" of the ground state. This identification is made clearer by the simpler statement

$$\langle 0|\hat{\sigma}_i^z|0\rangle = \pm N_0, \tag{1.15}$$

which also follows from the perturbation theory in g. We have $N_0 = 1$ for $g = 0$, but quantum fluctuations at small-g reduce N_0 to a smaller, but nonzero, value.

Now we make the simple observation that it is not possible for states that obey (1.11) and (1.14) to transform into each other analytically as a function of g. There must be a critical value $g = g_c$ at which the large $|x_i - x_j|$ limit of the two-point correlator changes from (1.11) to (1.14) – this is the position of the quantum phase transition, which is the

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focus of intensive study in this book. Our arguments so far do not exclude the possibility that there could be more than one critical point, but this is known not to happen for H_I, and we will assume here that there is only one critical point at $g = g_c$. For $g > g_c$ the ground state is, as noted earlier, a *quantum paramagnet*, and (1.11) is obeyed. We will find that as g approaches g_c from above, the correlation length, ξ, diverges as in (1.2). Precisely at $g = g_c$, neither (1.11) nor (1.14) is obeyed, and we find instead a power-law dependence on $|x_i - x_j|$ at large distances. The result (1.14) holds for all $g < g_c$, when the ground state is *magnetically ordered.* The spontaneous magnetization of the ground state, N_0, vanishes as a power law as g approaches g_c from below.

Finally, we make a comment about the excited states of H_I. In a finite lattice, there is necessarily a nonzero energy separating the ground state and the first excited state. However, this energy spacing can either remain finite or approach zero in the infinite lattice limit, the two cases being identified as having a gapped or gapless energy spectrum, respectively. We will find that there is an energy gap Δ that is nonzero for all $g \neq g_c$, but that it vanishes upon approaching g_c as in (1.1), producing a gapless spectrum at $g = g_c$.

1.4.2 Quantum rotor model

We turn to the somewhat less familiar quantum rotor models. Elementary quantum rotors do not exist in nature; rather, each quantum rotor is an effective quantum degree of freedom for the low-energy states of a small number of electrons or atoms. We will first define the quantum mechanics of a single rotor and then turn to the lattice quantum rotor model. The connection to the experimental models introduced in Section 1.3 is described below in Section 1.4.3. Further details of this connection appear in Chapters 9 and 19.

Each rotor can be visualized as a particle constrained to move on the surface of a (fictitious) $(N > 1)$-dimensional sphere. The orientation of each rotor is represented by an N-component unit vector $\hat{\mathbf{n}}_i$ which satisfies

$$\hat{\mathbf{n}}_i^2 = 1. \tag{1.16}$$

The caret on $\hat{\mathbf{n}}_i$ reminds us that the orientation of the rotor is a quantum mechanical operator, while i represents the site on which the rotor resides; we will shortly consider an infinite number of such rotors residing on the sites of a d-dimensional lattice. Each rotor has a momentum $\hat{\mathbf{p}}_i$, and the constraint (1.16) implies that this must be tangential to the surface of the N-dimensional sphere. The rotor position and momentum satisfy the usual commutation relations

$$[\hat{n}_\alpha, \hat{p}_\beta] = i\delta_{\alpha\beta} \tag{1.17}$$

on each site i; here $\alpha, \beta = 1 \ldots N$. (Here, and in the remainder of the book, we will always measure time in units in which

$$\hbar = 1, \tag{1.18}$$

unless stated explicitly otherwise. This is also a good point to note that we will also set Boltzmann's constant

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$$k_B = 1 \tag{1.19}$$

by absorbing it into the units of temperature, T.) We will actually find it more convenient to work with the $N(N-1)/2$ components of the rotor angular momentum

$$\hat{L}_{\alpha\beta} = \hat{n}_\alpha \hat{p}_\beta - \hat{n}_\beta \hat{p}_\alpha. \tag{1.20}$$

These operators are the generators of the group of rotations in N dimensions, denoted $\mathrm{O}(N)$. Their commutation relations follow straightforwardly from (1.17) and (1.20). The case $N = 3$ will be of particular interest to us. For this we define $\hat{L}_\alpha = (1/2)\epsilon_{\alpha\beta\gamma} L_{\beta\gamma}$ (where $\epsilon_{\alpha\beta\gamma}$ is a totally antisymmetric tensor with $\epsilon_{123} = 1$), and then the commutation relations between the operators on each site are

$$\begin{aligned} [\hat{L}_\alpha, \hat{L}_\beta] &= i\epsilon_{\alpha\beta\gamma}\hat{L}_\gamma, \\ [\hat{L}_\alpha, \hat{n}_\beta] &= i\epsilon_{\alpha\beta\gamma}\hat{n}_\gamma, \\ [\hat{n}_\alpha, \hat{n}_\beta] &= 0; \end{aligned} \tag{1.21}$$

the operators with different site labels all commute.

The dynamics of each rotor is governed simply by its kinetic energy term; interesting effects arise from potential energy terms that couple the rotors together, and these will be considered in a moment. Each rotor has the kinetic energy

$$H_K = \frac{J\tilde{g}}{2}\hat{\mathbf{L}}^2, \tag{1.22}$$

where $1/J\tilde{g}$ is the rotor moment of inertia (we have put a tilde over g as we wish to reserve g for a different coupling to be introduced below). The Hamiltonian H_K can be readily diagonalized for general values of N by well-known group theoretical methods. We quote the results for the physically important cases of $N = 2$ and 3. For $N = 2$ the eigenvalues are

$$J\tilde{g}\ell^2/2 \qquad \ell = 0, 1, 2, \ldots; \quad \text{degeneracy} = 2 - \delta_{\ell,0}. \tag{1.23}$$

Note that there is a nondegenerate ground state with $\ell = 0$, while all excited states are two-fold degenerate, corresponding to a left- or right-moving rotor. This spectrum will be important in the mapping to physical models to be discussed in Section 1.4.3. For $N = 3$, the eigenvalues of H_K are

$$J\tilde{g}\ell(\ell+1)/2 \qquad \ell = 0, 1, 2, \ldots; \quad \text{degeneracy} = 2\ell + 1, \tag{1.24}$$

corresponding to the familiar angular momentum states in three dimensions. These states can be viewed as representing the eigenstates of an even number of antiferromagnetically coupled Heisenberg spins, as discussed more explicitly in Section 1.4.3 and in Chapter 19, where we will see that there is a general and powerful correspondence between quantum antiferromagnets and $N = 3$ rotors.

We are ready to write down the full quantum rotor Hamiltonian, which will be the focus of intensive study in Parts II and III. We place a single quantum rotor on the sites, i, of a d-dimensional lattice, obeying the Hamiltonian

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$$H_R = \frac{J\tilde{g}}{2}\sum_i \hat{\mathbf{L}}_i^2 - J\sum_{\langle ij\rangle}\hat{\mathbf{n}}_i\cdot\hat{\mathbf{n}}_j. \tag{1.25}$$

We have augmented the sum of kinetic energies of each site with a coupling, J, between rotor orientations on neighboring sites. This coupling energy is minimized by the simple "magnetically ordered" state in which all the rotors are oriented in the same direction. In contrast, the rotor kinetic energy is minimized when the orientation of the rotor is maximally uncertain (by the uncertainty principle), and so the first term in H_R prefers a quantum paramagnetic state in which the rotors do not have a definite orientation (i.e. $\langle\mathbf{n}\rangle = 0$). Thus the roles of the two terms in H_R closely parallel those of the terms in the Ising model H_I. As in Section 1.4.1, for $\tilde{g} \gg 1$, when the kinetic energy dominates, we expect a quantum paramagnet in which, following (1.11),

$$\langle 0|\hat{\mathbf{n}}_i\cdot\hat{\mathbf{n}}_j|0\rangle \sim e^{-|x_i-x_j|/\xi}. \tag{1.26}$$

Similarly, for $\tilde{g} \ll 1$, when the coupling term dominates, we expect a magnetically ordered state in which, as in (1.14),

$$\lim_{|x_i-x_j|\to\infty}\langle 0|\hat{\mathbf{n}}_i\cdot\hat{\mathbf{n}}_j|0\rangle = N_0^2. \tag{1.27}$$

Finally, we can anticipate a second-order quantum phase transition between the two phases at $\tilde{g} = \tilde{g}_c$, and the behavior of N_0 and ξ upon approaching this point will be similar to that in the Ising case. These expectations turn out to be correct for $d > 1$, but we will see that they need some modifications for $d = 1$. In one dimension, we will show that $\tilde{g}_c = 0$ for $N \geq 3$, and so the ground state is a quantum paramagnetic state for all nonzero $\tilde{g}$. The case $N = 2$, $d = 1$ is special: there is a transition at a finite $\tilde{g}_c$, but the divergence of the correlation length does not obey (1.2) and the long-distance behavior of the correlation function $\tilde{g} < \tilde{g}_c$ differs from (1.27). This case will not be considered until Section 20.3 in Part IV.

1.4.3 Physical realizations of quantum rotors

We will consider the $N = 3$ quantum rotors first, and expose a simple and important connection between O(3) quantum rotor models and a certain class of "dimerized" antiferromagnets, of which $TlCuCl_3$ is the example we highlighted in Section 1.3. Actually the connection between rotor models and antiferromagnets is far more general than the present discussion may suggest, as we see later in Chapter 19. However, this discussion should enable the reader to gain an intuitive feeling for the physical interpretation of the degrees of freedom of the rotor model.

Consider a dimerized system of "Heisenberg spins" $\hat{\mathbf{S}}_{1i}$ and $\hat{\mathbf{S}}_{2i}$, where i now labels a pair of spins (a "dimer"). Their Hamiltonian is

$$H_d = K\sum_i \hat{\mathbf{S}}_{1i}\cdot\hat{\mathbf{S}}_{2i} + J\sum_{\langle ij\rangle}\left(\hat{\mathbf{S}}_{1i}\cdot\hat{\mathbf{S}}_{1j} + \hat{\mathbf{S}}_{2i}\cdot\hat{\mathbf{S}}_{2j}\right). \tag{1.28}$$

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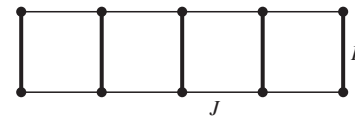

Fig. 1.4 A dimerized quantum spin system. Spins with angular momentum S reside on the circles, with antiferromagnetic exchange couplings as shown.

The $\hat{\mathbf{S}}_{ni}$ ($n = 1, 2$ labels the spins within a dimer) are spin operators usually representing the total spin of a set of electrons in some localized atomic states, see Fig. 1.4. On each site, the spins $\hat{\mathbf{S}}_{ni}$ obey the angular momentum commutation relations

$$\left[\hat{S}_\alpha, \hat{S}_\beta\right] = i\epsilon_{\alpha\beta\gamma}\hat{S}_\gamma \tag{1.29}$$

(the site index has been dropped above), while spin operators on different sites commute. These commutation relations are the same as those of the $\hat{\mathbf{L}}$ operators in (1.21). However, there is one crucial difference between the Hilbert spaces of states acted on by the quantum rotors and the Heisenberg spins. For the rotor models we allowed states with arbitrary total angular momentum ℓ on each site, as in (1.24), and so there were an infinite number of states on each site. For the present Heisenberg spins, however, we will only allow states with total spin S on each site, and we will permit S to be integer or half-integer. Thus there are precisely $2S + 1$ states on each site

$$|S, m\rangle \quad \text{with } m = -S \ldots S, \tag{1.30}$$

and the operator identity

$$\hat{\mathbf{S}}_{ni}^2 = S(S+1) \tag{1.31}$$

holds for each i and n. In addition to describing $TlCuCl_3$, Hamiltonians like H_d describe spin-ladder compounds in $d = 1$ [33, 102] and "double layer" antiferromagnets in the family of the high-temperature superconductors in $d = 2$ [129,322,337,443,444,506,507].

Let us examine the properties of H_d in the limit $K \gg J$. As a first approximation, we can neglect the J couplings entirely, and then H_d splits into decoupled pairs of sites, each with a strong antiferromagnetic coupling K between two spins. The Hamiltonian for each pair can be diagonalized by noting that $\mathbf{S}_{1i}$ and $\mathbf{S}_{2i}$ couple into states with total angular momentum $0 \leq \ell \leq 2S$, and so we obtain the eigenenergies

$$(K/2)(\ell(\ell+1) - 2S(S+1)), \qquad \text{degeneracy } 2\ell + 1. \tag{1.32}$$

Note that these energies and degeneracies are in one-to-one correspondence with those of a single quantum rotor in (1.24), apart from the difference that the upper restriction on ℓ being smaller than $2S$ is absent in the rotor model case. If one is interested primarily in low-energy properties, then it appears reasonable to represent each pair of spins by a quantum rotor.

We have seen that the $K/J \to \infty$ limit of H_d closely resembles the $\tilde{g} \to \infty$ limit of H_R. To first order in $\tilde{g}$, we can compare the matrix elements of the term proportional to J in H_R among the low-lying states, with those of the J term in H_d; it is not difficult to

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see that these matrix elements become equal to each other for an appropriate choice of couplings: see Exercise 6.1. Therefore we may conclude that the low-energy properties of the two models are closely related for large K/J and $\tilde{g}$. Somewhat different considerations in Chapter 19 will show that the correspondence also applies to the quantum critical point and to the magnetically ordered phase.

The main lesson of the above analysis is that the O(3) quantum rotor model represents the low-energy properties of quantum antiferromagnets of Heisenberg spins, with each rotor being an effective representation of a *pair* of antiferromagnetically coupled spins. The strong-coupling spectra clearly indicate the operator correspondence $\hat{\mathbf{L}}_i = \hat{\mathbf{S}}_{1i} + \hat{\mathbf{S}}_{2i}$, and so the rotor angular momentum represents the total angular momentum of the underlying spin system. Examination of matrix elements in the large-S limit shows that $\hat{\mathbf{n}}_i \propto \hat{\mathbf{S}}_{1i} - \hat{\mathbf{S}}_{2i}$: the rotor coordinate $\hat{\mathbf{n}}_i$ is the antiferromagnetic order parameter of the spin system. Magnetically ordered states of the rotor model with $\langle \hat{\mathbf{n}}_i \rangle \neq 0$, which we will encounter below, are therefore spin states with long-range antiferromagnetic order and have a vanishing total ferromagnetic moment. Quantum Heisenberg spin systems with a net ferromagnetic moment are *not* modeled by the quantum rotor model (11.1) – these will be studied in Section 19.2 by a different approach.

Let us now consider the $N = 2$ quantum rotors, and introduce their connection to the superfluid–insulator transition of bosons. For $N = 2$, it is useful to introduce an angular variable θ_i on each site, so that

$$\mathbf{n}_i = (\cos\theta_i, \sin\theta_i). \tag{1.33}$$

The rotor angular momentum has only one component, which can be represented in the Schrödinger picture as the differential operator

$$\hat{L}_i = \frac{1}{i}\frac{\partial}{\partial\theta_i} \tag{1.34}$$

acting on a wavefunction which depends on all the θ_i. The rotor Hamiltonian is therefore

$$H_R = -\frac{J\tilde{g}}{2}\sum_i \frac{\partial^2}{\partial\theta_i^2} - J\sum_{\langle ij\rangle}\cos(\theta_i - \theta_j), \tag{1.35}$$

which is a form that has appeared in numerous studies in different physical contexts. For $\tilde{g} \to \infty$, the eigenstates of H_R are of the form $\prod_i |m_i\rangle$, where m_i is the integer angular momentum quantum number of site i; in the Schrödinger form, these states have the wavefunction $\exp\left(i\sum_i m_i\theta_i\right)$. Now we interpret m_i as the *change* in occupation number of a boson trapped in a potential which has its minimum at site i. The boson could be an ultracold ^{87}Rb atom, or a Cooper pair in a superconducting quantum dot, as illustrated in Fig. 1.5. The occupation number is measured with respect to a "background" number of bosons found in the insulator, and hence m_i can take negative values whose absolute value does not exceed this number. In the rotor model, m_i can run all the way to $-\infty$, but as in the $N = 3$ case, we do not expect these additional high-energy states to be important for low-energy physics.

From the wavefunction of these localized boson states, we see that the term proportional to J in (1.35) has the effect of shifting nearest-neighbor pairs of angular momenta

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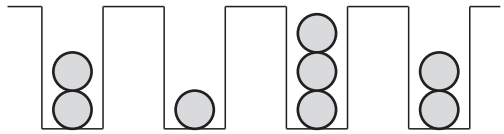

Fig. 1.5 Bosons hopping on a lattice with potential minima at site i. Relative to an insulator with 2 bosons on each site, the state shown has boson numbers $m_i = (\ldots 0, -1, 1, 0 \ldots)$.

as $m_i \to m_i \pm 1$, $m_j \to m_j \mp 1$. In other words, bosons tunnel between sites i and j with matrix element $-J$. Such a tunneling event becomes more probable when the optical lattice potential is weak (*i.e.* $\tilde{g}$ is small), and strong tunneling eventually induces a transition to the superfluid state where (1.27) is obeyed. We will see later in Chapter 8 why (1.27) implies superfluidity of the bosons. More details on the connection between $N = 2$ quantum rotors and quantum boson models appear in Chapter 9.

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